

KULACHI HANSRAJ MODEL SCHOOL
UNIT TEST - 1
SESSION: 2026-27
CAMBRIDGE CLASS VI (STAGE 7)
MATHEMATICS

MM: 20

NAME: _____ CLASS & SEC: _____ ROLL NO. : _____

INSTRUCTIONS:

- Answer all questions in the spaces provided on the question paper.
- You should show all your working on the question paper.

- 1) The sum of two integers is -9 and their product is -36.
Find the two integers.

[1]

-12 and 3

- 2) Work out $\sqrt[3]{125} - \sqrt[3]{27}$.

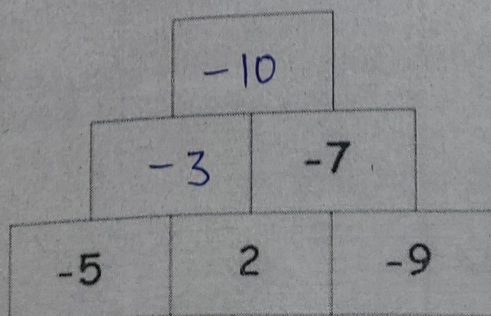
[2]

$$5 - 3$$

$$2$$

- 3) Use addition to complete the given number pyramid.

[1]



4) Show that $\sqrt{50}$ is closer to 7.

[1]

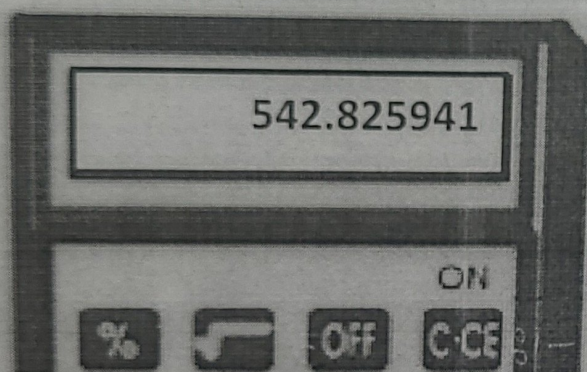
50 lies between 49 and 64

50 is closer to 49

$\sqrt{50}$ is closer to 7

5) Vani has worked out a calculation as shown on the calculator.

[1]



a) Round the answer to three decimal places. 542.826

b) Round the answer to five decimal places. 542.82594

6) Find the missing numbers.

[3]

a) $8 \times \boxed{-3} = -24$

b) $23 - (-47) = \boxed{70}$

c) $\boxed{-37} - 25 = -62$

d) $6 \times (-9 + 4) = \boxed{-30}$

e) $63.8 \times 10^4 = \boxed{638000}$

f) $43215 \div 10^7 = \boxed{0.0043215}$

7) a) Rashi says that 64 is a cube number but not a square number.

Shikha says that 64 is a square as well as a cube number.

[2]

i) Who do you agree with? Shikha

ii) Explain your answer.

$$64 = 4 \times 4 \times 4 = 4^3$$

$$64 = 8 \times 8 = 8^2$$

8) Estimate $-104.95 \div -15.24$ to the nearest whole number.

Show your working.

[2]

$$-105 \div -15$$

$$7$$

9) Shikha wants to simplify the fraction $\frac{42}{84}$ using their Highest Common Factor. Help her to do so.

Show your working.

[2]

HCF of 42 and 84 is 42

$$\frac{42 \div 42}{84 \div 42} = \frac{1}{2} = \frac{1}{2}$$

10) Find the lowest common multiple of 36 and 40.

Show your working.

[2]

$$\begin{array}{r|l} 2 & 36, 40 \\ \hline 2 & 18, 20 \\ \hline & 9, 10 \end{array}$$

$$\begin{aligned} \text{LCM} &= 2 \times 2 \times 9 \times 10 \\ &= 360 \end{aligned}$$

11) Use divisibility tests to check if 9, 7 and 11 are factors of 85261.

[3]

By 9

$$\text{Sum of digits} = 8 + 5 + 2 + 6 + 1 = 22$$

22 is not divisible by 9

So 85261 is not divisible by 9

By 7

$$\begin{array}{r} 85261 \\ - 2 \\ \hline 85249 \\ - 8 \\ \hline 8449 \\ - 8 \\ \hline 76 \end{array}$$

76 is not divisible by 7

So 85261 is not divisible by 7

By 11

$$\begin{aligned} \text{Sum of digits at} \\ \text{odd places} &= 1 + 2 + 8 \\ &= 11 \end{aligned}$$

$$\begin{aligned} \text{Sum of digits at} \\ \text{even places} &= 5 + 6 = 11 \end{aligned}$$

$$\begin{aligned} \text{Difference} &= 11 - 11 \\ &= 0 \end{aligned}$$

85261 is divisible by 11.